

Square Stability versus Hyperplane Stability in Banach Spaces

Audit of the supplied Cuntz-pair obstruction, literature status, and new reductions

Research report prepared from the supplied note and source paper

22 June 2026

Abstract

Albiac and Ansorena ask whether there exists a Banach space X such that $X \cong X \oplus X$ but $X \not\cong X \oplus \mathbb{F}$. The supplied note does not solve this problem; it proves a no-go statement for one attempted Gowers–Maurey spread construction. I verify that its conclusion is correct, identify a small definitional gap in the displayed proof, and give a proof that works even under the weaker definition written in the note. I also extend the obstruction from binary to every fixed finite arity and prove an address-rigidity result for doubly commuting Cuntz spread families.

A literature search through 22 June 2026 found no published or preprint proof or counterexample. In particular, the March 2026 source paper still states the problem as open, and a February 2026 paper still calls the hyperplane problem for the square-stable Kalton–Peck space Z_2 open. The report develops several additional reductions. The sharpest are: a K_0 characterization in terms of the rank-one idempotent; a reduction to the old Gowers–Maurey spaces S_m ; a conditional even-index theorem for Z_2 ; and a realification theorem saying that a square-stable complex space essentially incomparable with its conjugate would immediately yield a real counterexample. These observations materially narrow the problem, but they do not close the final gap. New deductions in this report have not been peer reviewed.

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1 Outcome

Let X be an infinite-dimensional Banach space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. The question is

$$X \cong X^2 := X \oplus X \quad \text{and} \quad X \not\cong X \oplus \mathbb{F}. \quad (1)$$

The first property will be called *square stability*, and the negation of the second is *hyperplane nonstability*.

The conclusions of this report are as follows.

- (1) The supplied finite-overlap obstruction is mathematically correct. As written, however, its proof uses closure of the spread family under products and adjoints, a clause present in the standard Gowers–Maurey definition but omitted from the note’s displayed definition. Theorem 3.1 repairs the proof without adding that clause.
- (2) The obstruction extends to exact d -ary Cuntz spread families for every fixed $d \geq 2$; see Theorem 3.3. It rules out an infinite reserve of fresh exact decompositions inside one proper family, but it does *not* settle the original single dyadic-tree strategy proposed by Gowers and Maurey.
- (3) No full answer was found in the literature. The problem was explicitly open in March 2026 [1]; the closely related Z_2 hyperplane problem was explicitly open in February 2026 [2].
- (4) Several rigorous routes reduce the problem to concrete operator-algebraic statements. The most promising currently appear to be the dyadic Gowers–Maurey index-transfer problem and triangularization of Fredholm operators on Z_2 .

Source crop: definitions and Question 2.15 in arXiv:2603.07324.

By Proposition 2.11, only c_0 , ℓ_1 , and ℓ_2 can be unconditional spreading models of these direct sums. By Proposition 2.12, ℓ_p and ℓ_q are the unique unconditional spreading models of $Z_{p,q}$, $p, q \in [1, \infty]$. In particular, any unconditional spreading model of $c_0(\ell_1)$, $\ell_1(c_0)$, $c_0(\ell_2)$ or $\ell_1(\ell_2)$ is c_0 , ℓ_1 or ℓ_2 .

Those who still expect nice universal structural results for Banach spaces would be tempted to conjecture that the answer to Question 2.8 is positive; however, as we will see in Section 5, this is not the case.

2.4. The hyperplane problem and related questions. Given a Banach space $\mathbb{X}$, there is $h = h(\mathbb{X}) \in \mathbb{N}_0$ such that

$$\{m \in \mathbb{N} : \mathbb{X} \oplus \mathbb{F}^m \simeq \mathbb{X}\} = \{hk : k \in \mathbb{N}\}$$

(see [7, Lemma 2.8]).

Definition 2.13. We say that a Banach space $\mathbb{X}$ is *hyperplane stable* if $h(\mathbb{X}) = 1$, i.e., $\mathbb{X} \oplus \mathbb{F} \simeq \mathbb{X}$. If $h(\mathbb{X}) > 0$, i.e., there is $h \in \mathbb{F}$ such that $\mathbb{X} \simeq \mathbb{X} \oplus \mathbb{F}^h$ we say that $\mathbb{X}$ is *hyperplane-sum stable*.

Similarly, there is $s = s(\mathbb{X})$ such that

$$\{m \in \mathbb{N} : \mathbb{X}^m \simeq \mathbb{X}\} = \{1 + sk : k \in \mathbb{N}\}.$$

Definition 2.14. A Banach space $\mathbb{X}$ is *square stable* if $s(\mathbb{X}) = 1$, i.e., $\mathbb{X}^2 \simeq \mathbb{X}$. If $s(\mathbb{X}) > 0$, i.e., there is $m \in \mathbb{N} \cap [2, \infty)$ such that $\mathbb{X}^m \simeq \mathbb{X}$, we say that $\mathbb{X}$ is *power stable*.

Most spaces that arise naturally in functional analysis are both hyperplane stable and square stable. The first counterexample to this paradigm goes back to [19], where James provided a Banach space $\mathbb{J}$ with $s(\mathbb{J}) = 0$ and $h(\mathbb{J}) = 1$.

The space $\mathbb{G}$ constructed by Gowers [15] witnesses that the answer to the Banach's hyperplane problem is negative, that is, there is a hyperplane nonstable Banach space. In fact, Gowers [15] proved that $h(\mathbb{G}) = 0$ and, later on, Gowers and Maurey [17] proved that $s(\mathbb{G}) = 0$. So, Gowers space $\mathbb{G}$ witnesses the existence of Banach spaces that are neither hyperplane stable nor square stable.

Gowers went on in [16] to build a Banach space $\mathbb{M}$ with $s(\mathbb{M}) = 2$. Subsequently, Gowers and Maurey [17] constructed for each $m \in \mathbb{N}$ a Banach space $\mathbb{S}_m$ with $s(\mathbb{S}_m) = m$, and another Banach space $\mathbb{H}_m$ with $h(\mathbb{H}_m) = m$. In the same paper, Gowers and Maurey conjectured that $h(\mathbb{S}_m) = 0$ for all $m \in \mathbb{N}$. To the best of our knowledge, this problem is still open. Similarly, we do not know whether the spaces $\mathbb{H}_m$, $m \in \mathbb{N}$, are square stable. In fact, it seems that the next question is still open.

Question 2.15. Is there a square stable Banach space failing to be hyperplane stable?

The Schröder–Bernstein's problem for Banach spaces asks whether, assuming $\mathbb{X}$ and $\mathbb{Y}$ are Banach spaces with $\mathbb{X} \trianglelefteq \mathbb{Y} \trianglelefteq \mathbb{X}$, then $\mathbb{X} \simeq \mathbb{Y}$.

This packet is a **strengthened partial/audit packet**. It supersedes the earlier June 14 packet, which recorded only the finite binary Cuntz-pair proper-set no-go. The previous version is preserved in a dated **history/** subfolder, and the submitted June 23 report is preserved in a dated **evidence/** subfolder. The main square-stable/hyperplane-nonstable question is still open.

2 Fredholm and K -theoretic reformulations

2.1 The index subgroup

For a Fredholm operator $T \in \mathcal{B}(X)$ use the convention

$$\text{ind } T = \dim \ker T - \text{codim}(TX; X).$$

The set of all such indices is an additive subgroup of $\mathbb{Z}$.

Proposition 2.1. *There is a unique $h(X) \in \mathbb{N}_0$ such that*

$$\{\text{ind } T : T \in \mathcal{B}(X) \text{ is Fredholm}\} = h(X)\mathbb{Z},$$

where $0\mathbb{Z} = \{0\}$. Moreover, for $n \geq 1$,

$$X \cong X \oplus \mathbb{F}^n \iff n \in h(X)\mathbb{N}.$$

In particular, X is hyperplane stable if and only if a Fredholm operator on X has index 1 (equivalently, index -1).

Proof. Fredholm indices add under composition, change sign on a parametrix, and include 0, so they form a subgroup of $\mathbb{Z}$. Every subgroup is $h\mathbb{Z}$. If T has index $n \geq 0$, choose complements to its kernel and range. The resulting finite-dimensional bookkeeping gives $X \cong X \oplus \mathbb{F}^n$. Conversely, if $J : X \rightarrow X \oplus \mathbb{F}^n$ is an isomorphism and $P : X \oplus \mathbb{F}^n \rightarrow X$ is the coordinate projection, then PJ is Fredholm of index n . $\square$

Thus the problem asks for a square-stable space whose index subgroup is either $\{0\}$ or $d\mathbb{Z}$ for some $d \geq 2$.

2.2 A sharp K_0 test

This subsection is over the complex field; an analogous statement can be made with real K -theory. Let $q \in \mathcal{B}(X)$ be any rank-one idempotent. Write $[q]_0, [I]_0 \in K_0(\mathcal{B}(X))$ for their classes.

Theorem 2.2 (rank-one criterion). *Assume that $X \cong X^2$. Then*

$$X \cong X \oplus \mathbb{C} \iff [q]_0 = 0 \text{ in } K_0(\mathcal{B}(X)).$$

Consequently, every complex solution of (1) must satisfy

$$[I]_0 = 0 \text{ but } [q]_0 \neq 0.$$

In particular, $K_0(\mathcal{B}(X))$ must be nontrivial even though its unit class vanishes.

Proof. If $X \cong X \oplus \mathbb{C}$, then $I \oplus q$ and I have isomorphic ranges, so $[q]_0 = 0$.

Conversely, $[q]_0 = 0$ means that, after stabilization, there is an idempotent $e \in M_n(\mathcal{B}(X))$ for which $q \oplus e$ is equivalent to e . Equivalence of idempotents in matrix algebras over $\mathcal{B}(X)$ is equivalence of their ranges. If $Y = \text{ran } e$, this gives

$$\mathbb{C} \oplus Y \cong Y.$$

The space Y is complemented in X^n . Square stability implies $X^n \cong X$, so, after transporting the projection, write $X \cong Y \oplus Z$. It follows that

$$X \oplus \mathbb{C} \cong (Y \oplus \mathbb{C}) \oplus Z \cong Y \oplus Z \cong X.$$

Finally, $X \cong X^2$ gives $[I]_0 = 2[I]_0$, hence $[I]_0 = 0$. □

Under square stability, the same argument gives

$$n[q]_0 = 0 \iff X \cong X \oplus \mathbb{C}^n.$$

Thus $[q]_0$ has order $h(X)$ when $h(X) > 0$, and infinite order when $h(X) = 0$. This agrees with Laustsen's Fredholm- K -theoretic framework [10].

Corollary 2.3. *Let X be a complex Banach space with $X \cong X^2$. If*

$$K_1(\text{Cal}(X)) = 0, \quad \text{Cal}(X) := \mathcal{B}(X)/\mathcal{K}(X),$$

then X is not isomorphic to a hyperplane.

Proof. The Fredholm index factors through the stable homotopy class of the invertible coset in $\text{Cal}(X)$, hence through $K_1(\text{Cal}(X))$. If this group is zero, every Fredholm operator has index zero. Apply Theorem 2.1. □

This gives a clean construction blueprint: realize square stability by an exact binary Cuntz family while forcing a Calkin algebra with trivial K_1 . The known constructions achieve these two requirements separately, but no known construction controls both simultaneously.

2.3 Structural restrictions on a counterexample

Proposition 2.4 (primary obstruction). *If X is primary and $X \cong X^2$, then $X \cong X \oplus \mathbb{F}$.*

Proof. The space $X \oplus \mathbb{F}$ is isomorphic to a complemented subspace of X^2 , hence of X . Write

$$X \cong (X \oplus \mathbb{F}) \oplus Z.$$

By primarity, either $X \oplus \mathbb{F} \cong X$, and there is nothing to prove, or $Z \cong X$. In the latter case

$$X \cong (X \oplus \mathbb{F}) \oplus X \cong X^2 \oplus \mathbb{F} \cong X \oplus \mathbb{F}.$$

□

Kalton proved stronger results for countably primary spaces with unconditional bases [9]. Thus a solution must be decidedly nonprimary.

Proposition 2.5 (shape of a hypothetical hyperplane). *Assume $X \cong X^2$ and let H be a hyperplane of X . Then*

$$H \cong X \oplus H.$$

Hence X and H complementably embed into one another. If $X \not\cong H$, they form a particularly rigid failure of the Banach-space Schroeder–Bernstein principle.

Proof. Pull H back through an isomorphism $X \oplus X \rightarrow X$. The inverse image is the kernel of a nonzero functional (f, g) on $X \oplus X$. If, say, $g \neq 0$, splitting $X = \ker g \oplus \mathbb{F}$ identifies this kernel with $X \oplus \ker g$. All hyperplanes of X are mutually isomorphic, so $\ker g \cong H$. $\square$

Lemma 2.6 (stable-reservoir obstruction). *If $X = Y \oplus Z$ and $Y \cong Y \oplus \mathbb{F}$, then $X \cong X \oplus \mathbb{F}$.*

Proof. $X \oplus \mathbb{F} \cong (Y \oplus \mathbb{F}) \oplus Z \cong Y \oplus Z = X$. $\square$

Consequently, a counterexample contains no complemented hyperplane-stable space. This eliminates the standard countable ℓ_p - and c_0 -sum stabilizations: they contain a complemented scalar ℓ_p or c_0 diagonal. It also explains why tensoring a hyperplane-nonstable space with a classical stable factor is counterproductive whenever that factor survives complementedly.

3 Audit and repair of the supplied spread argument

3.1 The omitted clause

For infinite $A = \{a_1 < a_2 < \cdots\}$ and $B = \{b_1 < b_2 < \cdots\}$, the Gowers–Maurey spread $S_{A,B}$ is defined by $S_{A,B}e_{a_k} = e_{b_k}$ and $S_{A,B}e_n = 0$ for $n \notin A$.

In the standard definition [7, Section 3], a proper family of spreads is

- (i) closed under composition and formal adjoints; and
- (ii) subject to the finite-overlap condition: for two distinct matrix positions, only finitely many family members are nonzero at both positions.

The supplied note displays only (ii). Its proof then asserts that $P_A = RR^*$ belongs to the family. That is automatic from (i), but not from the weaker displayed definition. Thus the proof is valid under the actual standard definition, while under the note’s literal definition it has a small gap.

The following proof removes the gap entirely.

3.2 A repaired binary theorem

Call a family *weakly proper* if it satisfies only the finite-overlap condition. An exact binary Cuntz spread pair is a pair (R_0, R_1) of spreads with

$$R_i^* R_j = \delta_{ij} I, \quad R_0 R_0^* + R_1 R_1^* = I.$$

Each R_i is therefore the unique spread from all of $\mathbb{N}$ onto an infinite set A_i , where $\mathbb{N} = A_0 \sqcup A_1$.

Theorem 3.1 (repaired supplied result). *A weakly proper family of spreads contains only finitely many exact binary Cuntz pairs.*

Proof. Suppose there are infinitely many pairs. Since a fixed unordered partition $\mathbb{N} = A_0 \sqcup A_1$ gives at most two labelled pairs, there are infinitely many distinct unordered binary partitions.

Choose three coordinates $a < b < c$. In each partition, two of these three coordinates lie in one cell. Passing to an infinite subfamily, a fixed pair, say a, b , lies in a cell B_m . For binary partitions the cells B_m may be assumed distinct: one cell determines its complement and hence the unordered partition.

Let R_m be the member of the corresponding Cuntz pair with range B_m . Define

$$r_m(t) = |B_m \cap \{1, \dots, t\}|.$$

Because R_m is the increasing spread from $\mathbb{N}$ onto B_m ,

$$R_m e_{r_m(a)} = e_a, \quad R_m e_{r_m(b)} = e_b.$$

There are only a possible values of $r_m(a)$ and only b possible values of $r_m(b)$. A second pigeonhole argument therefore gives an infinite subfamily on which both ranks are fixed. Infinitely many distinct spreads are then nonzero at the same two distinct matrix positions

$$(a, r(a)) \quad \text{and} \quad (b, r(b)),$$

contradicting weak properness. □

This proves that the conclusion of the supplied packet is right even with its weaker definition. The original projection proof is shorter if standard closure is restored.

3.3 Finite arity

The binary argument has a useful finite-arity extension.

Lemma 3.2 (partition lemma). *Fix $d \geq 2$. Let $\mathcal{P}$ be an infinite family of distinct partitions of $\mathbb{N}$ into d infinite cells. Then there exist distinct coordinates a, b and infinitely many distinct cells, drawn from partitions in $\mathcal{P}$, each containing both a and b .*

Proof. Induct on d . The case $d = 2$ is the three-coordinate argument used above. For general d , choose $d + 1$ coordinates. In every partition two lie in a common cell, so a fixed pair lies in a cell C_P for infinitely many partitions. If infinitely many C_P are distinct, we are done. Otherwise, pass to an infinite subfamily with one fixed cell C . Removing C leaves infinitely many distinct partitions of $\mathbb{N} \setminus C$ into $d - 1$ infinite cells. Apply the induction hypothesis. □

Theorem 3.3 (finite-arity no-go). *For every fixed $d \geq 2$, a weakly proper family of spreads contains only finitely many exact d -ary Cuntz spread families $(R_1, \dots, R_d)$ satisfying*

$$R_i^* R_j = \delta_{ij} I, \quad \sum_{i=1}^d R_i R_i^* = I.$$

Proof. Infinitely many such families give infinitely many distinct unordered d -cell partitions, up to the finite number $d!$ of labelings. Apply Theorem 3.2 to obtain a fixed pair a, b in infinitely many distinct range cells B_m . Select the corresponding spread $R_m : \mathbb{N} \rightarrow B_m$ and fix the two finite-valued ranks $|B_m \cap [1, a]|$ and $|B_m \cap [1, b]|$ on an infinite subfamily. This violates finite overlap exactly as in Theorem 3.1. □

3.4 Address rigidity and double commutation

Under the full standard definition, properness gives an additional obstruction.

Proposition 3.4 (address rigidity). *Let $\mathcal{S}$ be a standard proper family and let $(R_0, R_1) \subset \mathcal{S}$ be an exact binary Cuntz pair. For a word w in $\{0, 1\}$ put $R_w = R_{w_1} \cdots R_{w_n}$ and $P_w = R_w R_w^*$. The binary address map*

$$a : \mathbb{N} \longrightarrow \{0, 1\}^{\mathbb{N}}, \quad a(x)|_n = w \iff P_w e_x = e_x,$$

is injective.

Proof. At every level the projections $(P_w)_{|w|=n}$ are pairwise disjoint and sum to I , so every coordinate has a unique address. Along a branch the cylinder projections are distinct, since each cylinder splits into two nonzero child cylinders. If distinct coordinates x, y had the same address, then the infinitely many projections $P_{a(x)|_n} \in \mathcal{S}$ would all be nonzero at the two diagonal positions (x, x) and (y, y) , contrary to properness. $\square$

Corollary 3.5 (no second doubly commuting pair). *If a total spread $T \in \mathcal{S}$ commutes with every R_i and R_i^* , then $T = I$. Consequently, $\mathcal{S}$ cannot contain a second exact Cuntz pair that doubly commutes with the first one across generators.*

Proof. The commutation assumptions imply that T commutes with every P_w . If $T e_x = e_y$, then x and y belong to exactly the same cylinder ranges, so Theorem 3.4 gives $x = y$. Hence $T = I$. Two members of a second exact pair cannot both equal I . $\square$

Ordinary commutation without the adjoint relations is weaker and is not covered by this corollary.

4 What the literature currently gives

4.1 Status as of 22 June 2026

The source paper states the question explicitly and describes it as open [1, Question 2.15]. Searches were made for the exact question, for the phrases “isomorphic to its square” and “hyperplane”, for updates on the Gowers–Maurey S_m and H_m spaces, and for the hyperplane problem for Z_2 . No later proof or counterexample was found.

Two independent recent checkpoints reinforce that conclusion.

- Castillo, Gonzalez and Pino state that Z_2 is isomorphic to its codimension-two subspaces while its codimension-one problem remains unknown [3].
- Carrera and Ferenczi again call the Z_2 hyperplane problem open in a February 2026 preprint [2].

Since $Z_2 \cong Z_2^2$, a negative answer to its hyperplane problem would solve (1) immediately.

4.2 The two original Gowers–Maurey families

Gowers and Maurey constructed two relevant families [7]. For each $m \geq 2$ there is a space H_m whose finite-codimensional isomorphism classes are periodic modulo m ; in particular $h(H_m) = m$. Thus:

If any H_m is square stable, the problem is solved.

The square stability of these spaces is still unknown in the source paper.

They also constructed spaces S_m with periodic powers:

$$S_m^a \cong S_m^b \iff a \equiv b \pmod{m}.$$

Therefore $Y_m := S_m^m$ is automatically square stable, since $S_m^{2m} \cong S_m^m$.

Proposition 4.1 (reduction to S_m). *If S_m is not hyperplane stable for some $m \geq 2$, then S_m^m solves (1).*

Proof. Only hyperplane nonstability needs proof. Suppose S_m^m were hyperplane stable. Adding one copy of S_m gives

$$S_m^{m+1} \oplus \mathbb{F} \cong S_m^{m+1}.$$

But $S_m^{m+1} \cong S_m$ by periodicity, so $S_m \cong S_m \oplus \mathbb{F}$, a contradiction. $\square$

Gowers and Maurey conjectured the stronger assertion $h(S_m) = 0$. Thus their old conjecture would answer the 2026 question at once.

5 The single-tree Gowers–Maurey route

5.1 What the 1997 argument proves

In their ternary construction, Gowers and Maurey use right-append operators S_0, S_1, S_2 and left-prefix operators T_0, T_1, T_2 on the rooted ternary tree. They commute, satisfy

$$S_i^* S_j = T_i^* T_j = \delta_{ij} I, \quad \sum_i S_i S_i^* = \sum_i T_i T_i^* = I - P,$$

where P is the rank-one root projection. Modifying the right shifts at the root produces an exact ternary Cuntz family and hence a space $X \cong X^3$.

On an auxiliary ℓ_1 tree space they define

$$\Phi(U) = \sum_{i=0}^2 T_i U T_i^*.$$

For operators in the tree algebra, $\Phi(U) - U$ has finite rank, while $\Phi(U)$ is equivalent to three copies of U plus a root defect. This gives for a rectangular Fredholm matrix $U : Y^q \rightarrow Y^p$ the identity

$$2 \operatorname{ind} U = p - q.$$

In particular, every scalar Fredholm operator in the auxiliary algebra has index zero, and no rectangular Fredholm operator exists when $p - q$ is odd. This is the engine behind $X \cong X^3$ but $X \not\cong X^2$.

The authors then write that it is likely every Fredholm operator on the actual space has index zero and that a dyadic tree may yield a space isomorphic to its square but not to its hyperplanes [7, Section 4.4]. That sentence is essentially the present problem in embryonic form.

5.2 Why the supplied no-go does not stop this route

The supplied result excludes a countable reserve of fresh exact Cuntz spread pairs in one proper family. The 1997 proposal does not require such a reserve: it asks whether one dyadic family, together with the auxiliary tree algebra, controls all Fredholm indices. Thus the supplied theorem is a valid route obstruction, not a negative answer to the dyadic program.

5.3 The precise missing transfer

The Gowers–Maurey theorem yields a homomorphism from $\mathcal{B}(X)$ into a completion of the spread algebra modulo a seminorm-zero ideal. A further homomorphism lands in an auxiliary tree quotient. This is strong enough to detect the rectangular obstruction used to distinguish powers. It is not known to retain the finite-dimensional kernel/cokernel defect of a Fredholm endomorphism on X . In particular, finite-rank operators disappear in the quotient, exactly where the integer index is stored.

A complete dyadic proof would follow from any one of the following types of statement.

- (a) a natural morphism of extensions that sends a rank-one operator on X to the root rank-one operator in the auxiliary tree algebra;
- (b) injectivity, on the relevant K_1 classes, of the symbol map from the Calkin algebra of X to the auxiliary quotient; or
- (c) a direct amplification on X that is simultaneously a finite-rank (or inessential) perturbation of the identity on the whole approximating algebra and doubles Fredholm index.

The finite-overlap theorem explains why option (c) cannot be implemented by an infinite list of fresh exact spread pairs. Options (a) and (b) remain viable.

6 The Kalton–Peck candidate Z_2

6.1 Known operator structure

The Kalton–Peck space fits into a singular exact sequence

$$0 \longrightarrow \ell_2 \xrightarrow{i} Z_2 \xrightarrow{p} \ell_2 \longrightarrow 0.$$

It is isomorphic to its square and to its codimension-two subspaces. Recent work proves, among other things, that on Z_2 :

- strictly singular, strictly cosingular and inessential operators coincide;
- the product of two strictly singular operators is compact; and
- every semi-Fredholm operator has complemented kernel and range.

See [3].

An operator can be represented schematically by

$$T = \begin{pmatrix} \alpha & \beta \\ \delta & \gamma \end{pmatrix}.$$

If $\delta = 0$, the operator is upper triangular relative to the exact sequence; then $\alpha, \gamma \in \mathcal{B}(\ell_2)$ and

$$\alpha - \gamma \in \mathcal{K}(\ell_2).$$

This almost forces parity of the index.

6.2 A conditional parity theorem

Theorem 6.1 (upper-triangular parity). *Let $T \in \mathcal{B}(Z_2)$ be Fredholm. Suppose that T is an inessential (in particular, strictly singular) perturbation of an upper triangular operator*

$$U = \begin{pmatrix} \alpha & \beta \\ 0 & \gamma \end{pmatrix}.$$

Then

$$\operatorname{ind} T = 2 \operatorname{ind} \alpha,$$

and in particular $\operatorname{ind} T$ is even.

Proof. Inessential perturbations preserve Fredholmness and index, so it suffices to consider U . The upper triangular form gives a commutative morphism of the short exact sequence defining Z_2 , with α on the kernel and γ on the quotient. The algebraic snake sequence is

$$0 \rightarrow \ker \alpha \rightarrow \ker U \rightarrow \ker \gamma \rightarrow \operatorname{coker} \alpha \rightarrow \operatorname{coker} U \rightarrow \operatorname{coker} \gamma \rightarrow 0.$$

Since U is Fredholm, $\ker \alpha$ is finite-dimensional and $\operatorname{coker} \gamma$ is finite-dimensional. Hence γ is lower semi-Fredholm. The compact relation $\alpha - \gamma \in \mathcal{K}(\ell_2)$ implies that α is also lower semi-Fredholm; together with finite-dimensional kernel, α is Fredholm. Compact perturbation then makes γ Fredholm with $\operatorname{ind} \gamma = \operatorname{ind} \alpha$. Taking the alternating sum of dimensions in the snake sequence gives

$$\operatorname{ind} U = \operatorname{ind} \alpha + \operatorname{ind} \gamma = 2 \operatorname{ind} \alpha.$$

□

Corollary 6.2. *If every Fredholm operator on Z_2 is an inessential perturbation of an upper triangular operator, then Z_2 is square stable and has index subgroup $2\mathbb{Z}$; therefore it solves (1).*

Proof. The theorem excludes odd indices, while the known codimension-two isomorphism supplies an operator of index 2. □

6.3 The exact remaining corner

For every T , the lower-left entry can be written $\delta = pTi : \ell_2 \rightarrow \ell_2$. The quotient map p is strictly singular, so δ is strictly singular on ℓ_2 , hence compact. Compactness alone is not enough for the known triangularization theorem. Castillo–Gonzalez–Pino prove triangularization when δ has the stronger smoothing extension

$$\delta : \ell_f^* \longrightarrow \ell_f,$$

in their auxiliary notation. Proving this extra regularity for Fredholm T , or finding a weaker argument that extracts index parity directly from compact δ , would settle the problem through Z_2 .

7 A complex-conjugate route

A different route uses the parity forced by realification. Recall that complex spaces E and F are *essentially incomparable* when every operator in both cross directions is inessential.

Theorem 7.1 (realification parity). *Let E be a complex Banach space such that*

- (i) $E \cong E^2$ complex-linearly; and

(ii) E and its complex conjugate $\overline{E}$ are essentially incomparable.

Then the underlying real space $E_{\mathbb{R}}$ is square stable and every real Fredholm operator on it has even index. Hence $E_{\mathbb{R}}$ is not isomorphic to a real hyperplane.

Proof. Square stability survives restriction of scalars. Let T be a real Fredholm operator on $E_{\mathbb{R}}$. The complexification of $E_{\mathbb{R}}$ is complex-isomorphic to $E \oplus \overline{E}$. Relative to this decomposition, $T_{\mathbb{C}}$ has the form

$$T_{\mathbb{C}} = \begin{pmatrix} A & C \\ B & \overline{A} \end{pmatrix},$$

where $B : E \rightarrow \overline{E}$ and $C : \overline{E} \rightarrow E$. Essential incomparability makes the off-diagonal matrix inessential. Therefore $T_{\mathbb{C}}$ has the same index as $\text{diag}(A, \overline{A})$. The latter is Fredholm, so A and $\overline{A}$ are Fredholm and have equal complex index. Finally,

$$\text{ind}_{\mathbb{R}} T = \text{ind}_{\mathbb{C}} T_{\mathbb{C}} = \text{ind}_{\mathbb{C}} A + \text{ind}_{\mathbb{C}} \overline{A} = 2 \text{ind}_{\mathbb{C}} A.$$

The equality between the real index and the complexified index follows because the complexification of a finite-dimensional real kernel or cokernel has the same dimension over $\mathbb{C}$. $\square$

Corollary 7.2 (power-periodic version). *Suppose $E^{m+1} \cong E$ for some $m \geq 1$ and E is essentially incomparable with $\overline{E}$. Then the real space $(E^m)_{\mathbb{R}}$ solves (1).*

Proof. Finite powers preserve essential incomparability. When $m = 1$, the asserted square stability is exactly $E^2 \cong E$. When $m \geq 2$,

$$E^{2m} = E^{m-1} \oplus E^{m+1} \cong E^{m-1} \oplus E = E^m.$$

Apply Theorem 7.1 to E^m . $\square$

This theorem identifies two complementary pieces already present in the literature but not yet in one space.

- Kalton's explicit spaces Z_{α} are not isomorphic to their complex conjugates for $\alpha \neq 0$ [8], and Wenzel observed that they are isomorphic to their squares [11]. What is missing is essential incomparability; mere nonisomorphism is not enough.
- Ferenczi constructed complex structures whose conjugates are totally (and hence essentially) incomparable [4], but the underlying spaces are hereditarily indecomposable and therefore not square stable.

A construction combining the square periodicity of the first family with the cross-operator rigidity of the second would give a full answer by Theorem 7.1.

The theory of even real Banach spaces gives a related formulation. An even space admits a complex structure while its hyperplanes do not [5]. Thus any square-stable even space would solve the problem. Existing even examples are not known to have the required square stability; the real Z_2 is a prominent candidate [2].

8 Why several natural constructions fail

8.1 Finite-dimensional arithmetic is not cancellative

Let $[X]$ denote the isomorphism class of X in the commutative monoid of Banach spaces under direct sum. The desired equations are

$$[X] = 2[X], \quad [X] \neq [X] + 1.$$

There is no formal cancellation principle in this monoid. The equality $[X] = 2[X]$ by itself does not absorb the finite-dimensional class 1; this is exactly why elementary Eilenberg-swindle arguments stall.

8.2 Countable sums and tensor stabilizations

An ℓ_p -sum $(\bigoplus_n Y_n)_p$ with nonzero coordinates contains a complemented scalar ℓ_p diagonal by choosing one vector and one functional in every coordinate. The same holds for a c_0 -sum. By Theorem 2.6, such a space is hyperplane stable. Therefore taking countably many copies of a known hyperplane-nonstable space cannot solve the problem using a standard unconditional sum.

Likewise, if a tensor or vector-valued construction contains a complemented copy of a hyperplane-stable factor, the same lemma destroys the desired obstruction. A successful stabilization must be highly twisted: it must create $X \cong X^2$ without leaving any complemented stable reservoir.

8.3 An exact Cuntz pair is not itself an index-one operator

An exact pair gives an isomorphism $X^2 \cong X$, but a single branch is an infinite-codimensional range in the natural tree models. It does not supply a Fredholm shift with one-dimensional cokernel. To deduce hyperplane stability one would need a complemented one-vector-per-level branch or another finite defect mechanism. The Gowers–Maurey constructions are designed precisely to obstruct such complemented selections.

9 A focused research agenda

The preceding analysis leaves four concrete targets.

1. **Dyadic index transfer.** Construct a morphism of extensions, or a K_1 -injective symbol map, connecting the Fredholm index on the dyadic Gowers–Maurey space to the root-defect index in the auxiliary tree algebra. This would likely prove all Fredholm indices are zero.
2. **Z_2 regularity.** Prove that the compact corner $\delta = pTi$ of every Fredholm operator has enough smoothing to be removed by an inessential perturbation. Theorem 6.1 would then give index subgroup $2\mathbb{Z}$.
3. **Complex-conjugate fusion.** Produce a square- or power-periodic complex space essentially incomparable with its conjugate. The realification in Theorems 7.1 and 7.2 would solve the problem without further Banach-space decomposition arguments.
4. **K -theoretic realization.** Construct a square-stable X with $[I]_0 = 0$ and nonzero rank-one class $[q]_0$, or equivalently with trivial unit class but nontrivial finite-rank index obstruction. A realization with $K_1(\text{Cal}(X)) = 0$ and an exact lifted binary Cuntz family would be sufficient by Theorem 2.3.

10 Conclusion

The supplied note contains a valid and useful no-go theorem, after either restoring the standard closure clause in the definition of a proper spread family or using the repaired rank-pigeonhole proof above. Its method extends to every finite arity and yields an additional double-commutation rigidity phenomenon. It does not answer Albiac–Ansorena Question 2.15, and no answer was found in the literature through 22 June 2026.

The problem can now be phrased sharply in several equivalent or sufficient forms. In the complex case, a solution must have a vanishing unit class but a nonvanishing rank-one class in $K_0(\mathcal{B}(X))$. The old Gowers–Maurey spaces reduce the problem to their unresolved hyperplane conjecture. The Kalton–Peck space reduces it to Fredholm triangularization and an even-index theorem. The complex route reduces it to essential incomparability with the conjugate. Each of these is a concrete, falsifiable target; none is presently closed.

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